



ROBERT H. SMITH SCHOOL OF BUSINESS

Research study on ***Military Retirees in Maryland: Factors Influencing Residence Choice and Economic Impact Analysis***

Research Methods and Technical Notes

This portion of the research provides research methods and data analysis performed to identify factors influencing migration patterns of military retirees and the potential economic impact of tax exemptions for military retirement income in Maryland.

Literature Review: The research begins by examining existing literature on the policy's effects in other states. Studies from Arkansas, New Mexico, and Vermont highlight potential benefits like attracting retirees who contribute to the workforce and local economy. However, a study from Virginia emphasizes the need for a broader strategy beyond tax breaks to attract retirees.

Interviews and Public Data: Beyond tax policies, the study explores additional factors influencing military retiree relocation decisions. Interviews were conducted with key figures in Maryland's Department of Veterans Affairs (MDVA). Additionally, publicly available resources like the MDVA's economic reports and surveys of military families provided further insights. Given these limitations, the study recommends further research using individual-level data to create more precise models. Additionally, exploring the impact on specific demographics within the military retiree population could yield valuable insights.

Limitations and Methodology: The study acknowledges limitations in accessing individual-level data on migration patterns. State-level data is used, with mitigation strategies like careful model selection and data validation to address potential bias .

- **Factors Influencing Migration:** Regression analysis revealed factors like the presence of active-duty personnel, construction jobs, government jobs, and military installations as significant factors attracting military retirees.
- **Migration Analysis:** An analysis of migration patterns nationwide and within Maryland was conducted using data from various sources like the Census Bureau and the IRS.
- **Net Fiscal Impact:** The study modeled the net economic impact of tax exemptions. While initial revenue loss is projected, the model suggests a long-term positive impact on state revenue due to factors like increased spending by retirees, sales tax, and property taxes.

LITERATURE SURVEY AND INTERVIEWS

A. Literature Surveys

There were several state-specific studies considered on the effect of taxes on the veteran population per state. These states were Arkansas, New Mexico, Vermont, and Virginia.

1. Arkansas

The Arkansas study¹ discussed the potential benefits and drawbacks for states that exempt military retirement income from state taxes. In terms of benefits, it highlighted that attracting military retirees could inject a steady source of income into the state, reduce the burden on public services since military retirees typically require fewer state services (e.g., schools) as their children are likely grown, and contribute to workforce development because many retirees have valuable skills and may seek second careers. The study recognized the cost to the state as the loss of tax revenue from the exempted income. In its conclusion, the simulations suggested that the policy could benefit the state budget in the long run, with the timeframe for positive impact depending on the migration rate of retirees, effective income tax rate, and retirees' post-retirement earnings, indicating that exempting military retirement income may be beneficial but dependent on several factors.

2. New Mexico

The New Mexico study² considered offering tax exemptions to military retirees to make the state a more attractive place to settle down. While the state would lose some income tax revenue from exempting military retirement income, this cost was likely outweighed by the gains. Military retirees bring in spending money that boosts sales tax, potentially buy homes that increase property tax revenue, and if they have other income sources, contribute to income tax revenue as well. Additionally, these retirees often have valuable skills and experience that benefit the workforce, and they typically require fewer state services because their children are likely grown. This study resulted in the passing of up to \$30,000 income tax exemption for military retirees and surviving spouses.

3. Vermont

The Vermont legislative report³ considered eliminating the tax on military retirement income. The report hypothesized that exempting this tax could attract more military retirees to the state who are often still of working age with valuable skills, which would help Vermont's labor shortage. Additionally, the report believed that these retirees would bring in extra income through spending and potentially starting second careers, further boosting the economy. This report supported Vermont's initiative, which led to the passing of legislation providing a \$10,000 income exemption for those with an annual income of \$50,000 or less.

4. Virginia

The Virginia study⁴ concluded that while tax breaks can be enticing, simply offering an exemption for military retirement income might not be enough to attract a significant number of retirees to Virginia and offset the lost tax revenue. The national trend, and even the state's own trend, showed a decrease in the retiree population. Virginia would likely need to rely on attracting retirees from other states rather than just encouraging current residents to stay. To achieve this, an aggressive marketing campaign highlighting job opportunities, good schools, and a high quality of life would be crucial.

¹ Hamilton, Gregory. University of Arkansas at Little Rock. "Analysis of the Fiscal Impacts Associated with Exempting Federal Military Retiree's Pensions from Arkansas State Personnel Income Tax". February 12, 2015.

² Military / Veterans Leadership Council. "Military Pay Tax Exemption". <https://www.nmlegis.gov/>. September 14, 2021.

³ Elredge, Matt. "The Benefits of Eliminating the Tax on Military Retirement Income in Vermont". <https://legislature.vermont.gov/>. February 2022.

⁴ Nodjoman, Sal & Schwalb, Matt. Matrix Design Group. "Fiscal Impacts of Retired Military Service Personnel on the Commonwealth of Virginia". March 16, 2017.

Based on the literature reviews mentioned above, this research examined the military retirement pay tax policies across all 50 U.S. states and Washington D.C. We found three states that fully tax military retirement income. Eleven states had established specific calculations defined by state legislation to determine the tax on this income. The remaining states did not tax military retirement pay at all. (Please see Appendix A for a visualization of the State Tax Comparison Map for Military Retirement Pay.)

B. Interviews and Other Third-Party Surveys

In addition to examining tax policies, this research explored other factors influencing military retirees' choice of retirement state. Interviews were conducted with key figures: Anthony Woods (Maryland Department of Veterans Affairs Secretary), Andrew Libraty (MDVA Deputy Chief of Staff), Morgan Murphy (MDVA Director of Intergovernmental & Legislative Affairs), and Emmett Roberts (Maryland Veterans Chamber of Commerce Founder and President). Publicly available resources from the MDVA, including the "Overview of the Economic Conditions in Maryland"⁵ and the MDVA FY 2025 Operating Budget, and surveys conducted by Blue Star Families capturing the experience of over 7,400 military family respondents⁶, were also analyzed to identify initial relevant factors. Based on these additional references, the following major categories were considered in the analysis: demographics, income-related factors, education system, economic indicators, crime rate, and veterans affairs and military infrastructures located in the state.

RESEARCH LIMITATIONS

A. Factors Influencing Migration Patterns of Military Retirees

One of the objectives of the research was to investigate the factors influencing the migration patterns of military retirees across various states in the United States. While numerous studies had focused on the tax implications for this demographic, there was a notable gap in understanding the specific decision-making criteria employed by these individuals. Given the constraints in accessing individual-level data, this study utilized state-level data to analyze these migration patterns.

However, this approach presented potential limitations. State-level data might not have captured the nuanced preferences and behaviors of individual military retirees, and individual preferences could vary significantly, potentially obscuring variations when aggregated. Additionally, there was a risk of drawing incorrect conclusions about individual behaviors based on state-level trends, known as the ecological fallacy, which could lead to misleading inferences that did not accurately represent the motivations and decisions of individual retirees.

To address these limitations, the study incorporated several mitigation strategies. Firstly, there was a careful evaluation of the underlying relationships between variables used in the model, ensuring that assumptions made about these relationships were reasonable and well-supported by existing literature. The model was selected with an appropriate level of complexity to avoid overfitting the state-level data, ensuring that it remained generalizable and could offer insights applicable to individual-level behaviors. Additionally, the study prioritized the use of actual data whenever available, particularly from reliable sources such as the Maryland state website and other government websites, to enhance the validity and reliability of the findings.

The study utilized data from various reliable sources, including state tax revenue data, state demographic data, and state-level economic indicators. The research employed a thorough analytical approach, encompassing descriptive analysis to summarize key characteristics of the data and identify trends and patterns, and inferential analysis to infer relationships between state-level factors and migration patterns while accounting for potential confounding variables.

⁵ Maryland Department of Labor. <https://mwejobs.maryland.gov/>. January 2024.

⁶ Blue Star Families. Military Lifestyle Survey. <https://bluestarfam.org/research/mfls-survey-release-2024/>. Accessed July 2024.

B. Net Fiscal Impact Model

Ideally, the most accurate data on actual tax revenue loss from a potential increase in military retirement exemptions would come from the Maryland Comptroller's Office. This data would allow the research to precisely calculate the impact based on existing military retiree tax filers, identifying their tax brackets and other demographic profiles. However, for this study, such detailed information was not readily available. Instead, the study relied on publicly available data about the number of exemption applicants and estimated costs associated with full exemption as indicated in Maryland Senate Bill 346 (Keep Our Heroes Home Act)⁶ to build an estimated tax effect model. This model estimated the marginal dollar impact for every dollar of retirement income exemption and for every new military retiree who decided to reside in Maryland. As the research also attempted to estimate 5-year and 10-year tax policy implications, data from the Department of Defense and model assumptions identified from various federal government websites were used whenever available.

METHODOLOGY AND RESEARCH APPROACH

A. OLS Regression Analysis on Factors Influencing Military Retiree Migration Patterns

This research utilized the OLS (Ordinary Least Squares) multiple linear regression methodology to identify the relationships between military retirees and various state-level factors. A variety of publicly available data sources were used to construct the database for the models. The dataset included factors potentially influencing military retirees' decisions on which state to choose for migration or retirement for the years 2020, 2021, and 2022, with data aggregated at the state level.

The dataset compiled a variety of variables from multiple sources to analyze the factors influencing the number of military retirees residing in different states in the US. The primary variable of interest was the number of military retirees per state from 2020-2022, sourced from the Department of Defense Actuary⁷. Other demographic and economic variables included the total population and total number of veterans per state, provided by the United States Census Bureau⁸. Tax-related variables included the classification of states based on the exemption of military retirement tax and the percentage exemption, gathered from various state websites and calculations. Cost-of-living data for 2024 was sourced from the Economic Policy Institute's Family Budget Calculator⁹. Economic indicators such as real per capita personal income and regional price parities were obtained from the U.S. Bureau of Economic Analysis¹⁰. Crime rates, state tax burden, and the number of veteran healthcare facilities were sourced from the National Incident-Based Reporting System¹¹, the Tax Foundation¹², and the Department of Veterans Affairs¹³, respectively. Education-related variables included the number of operating public school districts, public school expenditure per student, and National Assessment of Educational Progress (NAEP) assessment scores in math and reading, all from the National Center for Education Statistics¹⁴. The dataset also included data on population density, number of active-duty personnel, and military installations, sourced from the US Census Bureau and the Department of Defense¹⁵. Employment data across various sectors, such as construction, manufacturing, transportation, finance, real estate, healthcare, government, and others, were provided by the Bureau of Economic Analysis. See Appendix B for the detailed explanation of the dataset.

⁶ Maryland Senate Bill 346. Income Tax - Subtraction Modification - Military Retirement Income (Keep Our Heroes Home Act). <https://mgaleg.maryland.gov/mgawebsite/Legislation/Details/sb0346?ys=2024RS>. Accessed July 2024.

⁷ Office of the Actuary US Department of Defense. <https://actuary.defense.gov/>. Accessed July 2024.

⁸ United States Census Bureau. <https://data.census.gov/>. Accessed July 2024.

⁹ Family Budget Calculator. Economic Policy Institute. <https://www.epi.org/resources/budget/>. Accessed July 2024.

¹⁰ US Bureau of Economic Analysis. <https://www.bea.gov/>. Accessed July 2024.

¹¹ National Incident-Based Reporting System (NIBRS) Tables. <https://catalog.data.gov/dataset/?tags=nibrs>. Accessed July 2024.

¹² Walczak, Jared. Tax Foundation. How High Are State and Local Tax Collections in Your State?. <https://taxfoundation.org/data/all/state/state-local-tax-collections-per-capita-fy-2021/>. November 21, 2023.

¹³ US Department of Veterans Affairs. <https://www.data.va.gov/>. Accessed July 2024.

¹⁴ U.S. Department of Education, Institute of Education Sciences, National Center for Education Statistics. 2022 Mathematics and Reading Assessment Profiles. <https://www.nationsreportcard.gov/>. Accessed July 2024.

¹⁵ Defense Manpower Data Center US Department of Defense. <https://dwp.dmdc.osd.mil/dwp/app/dod-data-reports/workforce-reports>. Accessed July 2024.

The dependent variable in the model was the number of military retirees residing in a state. The independent variables or predictors included the number of active-duty military personnel present in a state to account for the military-civilian gap, tax variables to capture tax exemptions available in each state, and the number of military installations. Additionally, the model considered the number of operating public school districts and state reading and math scores to reflect the quality of education in each state. It also included various job opportunities per industry available in the state, variables explaining income and cost of living, and several variables that describe climatic conditions in the state.

Regression analysis revealed that the following factors affected military retirees' decisions to choose a particular state, in order of preference/importance:

- Number of active-duty personnel
- Construction jobs
- Government jobs
- Number of military installations
- Real estate and rental leasing jobs
- Population density
- Real per capita personal income
- Retirement tax exemptions
- Regional price parities

See Appendix C for the factors analysis regression results and technical summary.

B. Migration Flow Analysis

Data for this migration analysis was sourced from the US Census Bureau¹⁶, Internal Revenue Service¹⁷, and Department of Defense¹⁸. One of the analysis encompassed all US states, including Washington D.C., focusing on the period between 2010 and 2022. Given the limited availability of specific military retiree population data, migration patterns from states with high in-migration to Maryland were analyzed as a proxy. This approach enabled the identification of overall migration trends and potential shifts in recent years between states. (See Appendix D for the general population migration flow.) To visualize trends, military retiree population data for Maryland from 2010 to 2022 was graphed. Additionally, Maryland's military retiree growth rate was benchmarked against the national average for the same period to assess relative growth. (See Appendix E for military retiree population trends and Appendix F for regional military retiree population movements post-pandemic.)

C. Net Fiscal Impact Analysis

The net economic impact model was calculated by following a three-step process:

1. Estimate the amount of income tax revenue forgone for each dollar increase in exempt military retirement income.
2. Possible revenue sources to offset foregone income:
 - a. Assessing new retiree impact: Estimating the increased income tax revenue from new military retirees moving to Maryland (newly retired or from other states)
 - b. Existing retirees in Maryland potentially earning higher salaries
3. Estimate the 5-year and 10-year breakeven forecasts.

By analyzing these marginal impacts, the study evaluated the net economic effect on a per-dollar basis and/or per person basis. To estimate the overall state-level impact, the model utilizes projections from Senate Bill 346 as well as an attempt on the forecasts using data from DOD's Office of the Actuary Retirement Payments.

¹⁶ United States Census Bureau. <https://data.census.gov/>. Accessed July 2024.

¹⁷ IRS US Population Migration Data. <https://www.irs.gov/statistics/soi-tax-stats-migration-data>. Accessed July 2024.

¹⁸ Office of the Actuary US Department of Defense. <https://actuary.defense.gov/>. Accessed July 2024.

1. Quantifying forgone income tax revenue for every dollar of exemption

The biggest factor affecting estimated lost income tax was the state income tax bracket¹⁹ each military retiree fell into. To account for this uncertainty, the study used a discrete bounded distribution based on Maryland's existing tax brackets. Another unknown factor was the local tax rate, which varied by county. While the actual rates ranged from 2.25% to 3.2%, the study assumed the highest rate (3.2%) due to limited data on military retiree distribution by county. Notably, four of Maryland's most populous counties (Montgomery, Prince George's, Baltimore County, and Baltimore City) already had the 3.2% rate. This study focused on marginal impacts, estimating the lost income tax revenue for each dollar of exemption applied to each military retiree filing taxes in Maryland. (Details on calculating the marginal foregone income tax are provided in Appendix G.) Nevertheless, the precise figures for state and local tax revenue from the Comptroller's Office, both current and projected, were deemed more accurate than these estimates. This preliminary analysis was conducted to provide a rough approximation for potential use in future policy studies.

While increased disposable income for military retirees due to tax savings could potentially reduce the state's foregone income tax revenue, the marginal estimates for this section did not account for this effect. Predicting how retirees would spend their additional income was uncertain, as it could be spent outside Maryland, on tax-exempt items, or saved. Therefore, any potential increase in sales tax revenue from higher disposable income, while beneficial, was not included in this section.

2. Estimating possible revenue sources to offset foregone income

New Military Retiree Migrant to Maryland

This study also analyzed the estimated potential income of military retirees who choose Maryland as their retirement destination. The key factor influencing their income is their post-retirement career choice. While military rank, age, and education play a role, the industry sector has been identified as the most significant predictor based on existing research. Our analysis focuses on the construction, real estate, government, and defense contracting industries, which are commonly pursued by military retirees according to the study's regression on migration predictors for this group. Salary estimates for these industries in Maryland were obtained from the U.S. Bureau of Labor Statistics (BLS) May 2023 State Occupational Employment and Wage Estimates²⁰ and Zip Recruiter²¹. The research primarily focused on income tax revenue and did not consider other potential revenue streams such as sales tax, property tax, or business tax from these new residents. While these additional benefits are positive, they would require further analysis and assumptions and are therefore not quantified here. These new migrants can come from recently retired military personnel fresh out of service or from military retirees who may relocate from other states. Appendix H provides a breakdown of the descriptive summary statistics for select jobs' household income, total income tax, and state income tax for military retirees' households, considering the typical jobs military retirees pursue, as well as their marital status and the employment status of their spouse.

Existing Military Retiree in Maryland

In this section of the study estimates the minimum income increase required for existing retirees to offset the lost income tax revenue resulting from an increased tax exemption. A sensitivity analysis table was created to demonstrate this impact. However, the specific tax bracket for existing retirees who experience a salary increase is uncertain, meaning the breakeven point for each dollar of exemption varies by tax bracket (see Appendix I). Due to data limitations, the probability of each of the 64 potential scenarios cannot be calculated. Nonetheless, this table may remain valuable for future policy research on military retirees.

¹⁹ Maryland Income Tax Rates and Brackets. [Tax Rates \(marylandtaxes.gov\)](https://www.marylandtaxes.gov/). Accessed July 2024.

²⁰ May 2023 State Occupational Employment and Wage Estimates for Maryland. US Bureau of Labor Statistics. [Maryland - May 2023 OEWS State Occupational Employment and Wage Estimates \(bls.gov\)](https://www.bls.gov/publications/may2023/maryland.htm). Accessed July 2024.

²¹ Department of Defense Contractor Salary in Maryland. Zip Recruiter. <https://www.ziprecruiter.com/Salaries/Dod-Contractor-Salary--in-Maryland#TableView>. Accessed July 2024.

Furthermore, a Monte Carlo simulation was also conducted to estimate how much additional income each military retiree already living in Maryland would need to earn to cover the estimated foregone total income tax revenue per person. See Appendix J for the results of the simulated breakeven household income needed to cover \$61.3 million (from SB 349) using the population of existing military retirees in Maryland, which is 55.5K (from DoD 2022 military retirees in Maryland). This resulted in an estimated foregone income tax per household for full exemption of \$1100.

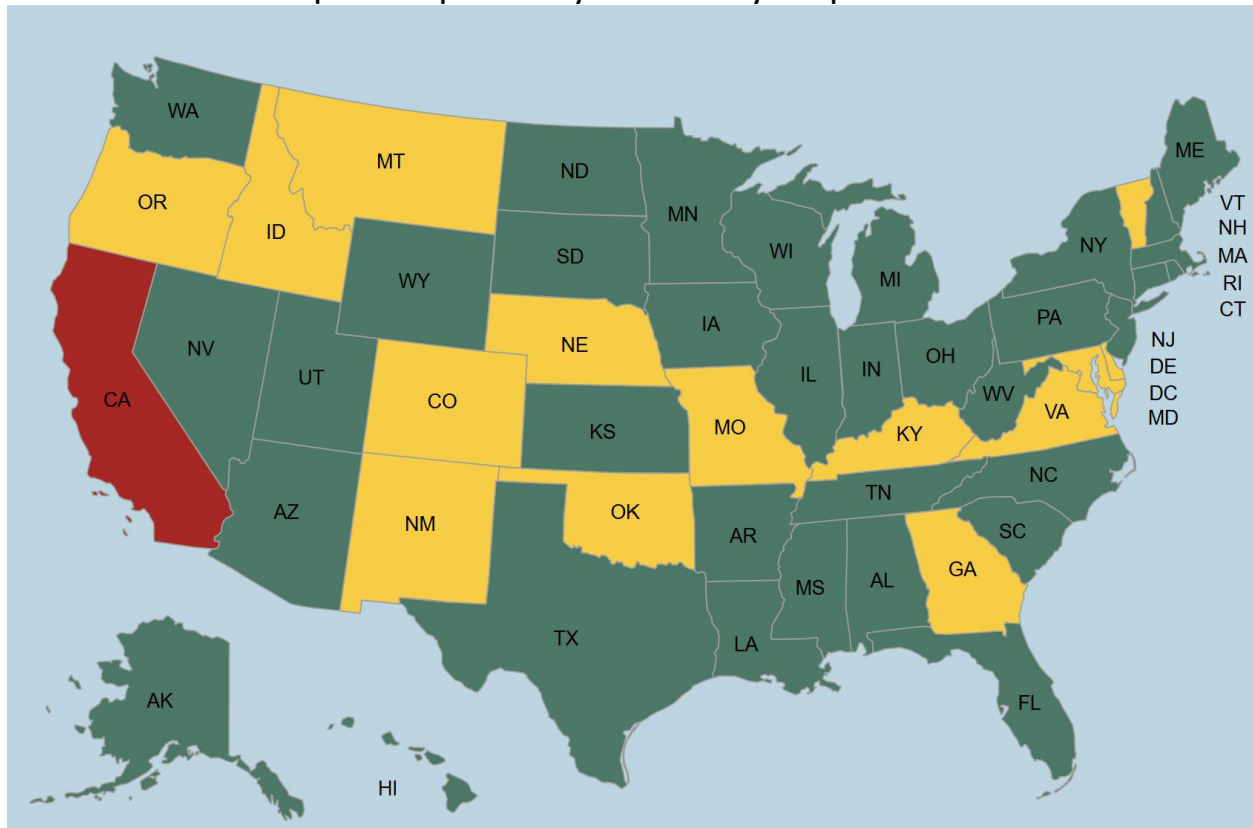
3. Projecting 5-year and 10-year breakeven forecasts

Lastly, the research has attempted to estimate net income tax revenue for Maryland, considering tax revenues from income, property, sales, and gas due to new military migration to the state. It should be noted that the forecast used DoD non-disability retirement payments as of 2022 to estimate the forgone tax. The tax exemption initially results in a significant revenue loss for the state, with negative impacts observed for the first nine years. The situation starts to improve by year 10, where the net impact is neutral. From year 11 onwards, the state begins to experience positive revenue gains, with substantial increases in the last few years of the analysis period. Overall, while the tax exemption poses short-term financial challenges, it is projected to deliver significant long-term financial benefits, surpassing the initial losses by the end of the 15-year period. See Appendix K for visualization net tax revenue forecasts for 15 years.

The net revenue forecasts were based on several key assumptions. It was estimated that the average retirement pay for military retirees in Maryland is \$55,686 for officers and \$26,040 for enlisted personnel. The total foregone income tax revenue was projected to be \$67,044,283, with \$26,773,422 from officers and \$40,270,861 from enlisted personnel. Additional revenue streams considered included sales tax, estimated at a 6% tax rate based on a per capita personal consumption expenditure of \$29,158, and gasoline tax, estimated at a 14.5% tax rate with expenditures at \$1,401. Property tax revenue was based on an average assessed value of \$268,443 with a 1.1% tax rate, assuming 75% of military retirees are homeowners. Household income estimates were adjusted from the US mean household income of \$106.4K to account for family/marital status, educational attainment, house rental status, Mid-Atlantic region premium, and age. Maryland's median household income is \$95K, but the estimated household income for military retirees is projected to be higher, particularly for officers, due to their likely experience and premium on their labor contribution. The estimated household incomes vary by age and rank, with higher earnings for officers across all age brackets compared to enlisted personnel.

Additional revenue streams such as sales tax, gasoline tax, and property tax were considered in intermediate calculations to provide a comprehensive view of the financial impact. This multifaceted approach supports the conclusion that, despite initial fiscal challenges, the tax exemption will ultimately be beneficial for the state's revenue in the long run.

APPENDIX A. State Tax Comparison Map for Military Retirement Pay Exemption



Source: Military Officers Association of America Military State report Card and Tax Guide <https://www.moaa.org/content/state-report-card/statereportcard/>. Accessed July 2024.

Map Legend:

- Green: Military retirement pay is fully exempt from state taxation.
- Yellow: Military retirement pay receives partial exemption from state taxation.
- Red: Military retirement pay is exempt from state taxation

Updates:

- Along with California, Oregon (with exemption if served before October 1, 1991) and D.C. (with some exemption for 62 years old and up) tax military retirement pay according to federal guidelines.
- Military retirement pay in Nebraska and Oklahoma is fully exempt from state taxation.

APPENDIX B. Dataset Construction

Variables

a. Demographics: This includes number of military retirees, total veteran population and veteran population by gender, number of military personnel receiving Concurrent Retirement and Disability Pay (CRDP) and Combat-Related Special Compensation (CRSC) payments and the number of active-duty military personnel per state and the population density which represents average number of people per square miles.

b. Tax Related: This includes exempting Military Retirement Tax which is a factor variable where full exemption is populated as 0, taxable is populated as 1 and partial exemption is populated as 2 and retirement tax exemption column which is Calculated as Exemption per State (\$) over Federal Average Non-Disability Military Retirement Pay: \$18K (2020), \$19K (2021), and \$21K (2022). Additionally estimated annual state income tax variable is calculated considering the average income of military retiree to be \$197K.

c. Jobs: This includes the number of jobs available in each state, further categorized into various types of jobs. This breakdown helps to account for the impact that specific job types may have on a military retiree's decision to retire in a particular state. The variables under this category are total number of jobs available, number of wage and salary employment, and jobs in the field of construction, manufacturing, transportation and warehousing, finance and insurance, real estate and rental and leasing, health care and social assistance, government and government enterprises, forestry fishing and related activities, information, educational services, farm proprietors, non-farm proprietors, farm and non-farm employment.

d. Income Related: This includes the variables related to income that is or may be received by the military retiree. Variables are average annual pension received by the veterans by state, CRDP and CRSC monthly payment received by the military retirees, real per capita personal income to capture the measure of purchasing power in each state.

e. Educational Infrastructure: This includes variables for the state math score and state reading score, which represent the Grade 8 assessments in Mathematics and Reading associated with the National Assessment of Educational Progress (NAEP). These variables capture the educational progress in each state. Variables in this category also include the number of operational public-school districts and public-school expenditure per student to capture the educational services and infrastructure within public schools, influencing educational quality and outcomes.

f. Economic Indicators: This includes variables Annual Cost of Living which measures the yearly expenses in USD for a household of two adults without children, using data from either the capital city's metro area or the largest metro area if capital data is unavailable and the Regional Price Parities (RPP) which indicate relative price levels compared to the national average (base of 100 for the USA), reflecting cost-of-living variations across regions based on average prices of goods and services. Additionally, it includes ease of doing business score variable which is the average ease of doing business score per state.

g. Crime Rate: This variable encompasses the crime rate, calculated by dividing the total number of offenses reported by participating agencies by the population they cover. Total offenses comprise crimes against persons, crimes against property, and crimes against society. This also includes variables average violent crime offenses and average property crime offenses which are calculated as average no of violent crimes per 100,000 people for a 10-year period and average no of property crimes per 100,000 people for a 10-year period respectively.

h. Veteran Affairs and Military Infrastructure Metrics: This includes number of veteran healthcare facilities which is determined as the sum of total outpatient care facilities, vet centers (counseling centers), inpatient care sites, Veterans Benefits Administration Regional Office, and national & state cemeteries for veterans per state and the number of military installations which is the count of total number of approved locations per state that are operated by individual branches of the military (e.g., Army, Navy, Air Force). It indicates the presence and distribution of

military bases, installations, and facilities within each state, which are essential for military operations, training, and support activities.

g. Climate Variables: This includes the extreme weather events variable tracking the frequency of severe weather incidents like hurricanes and storms in a state over the past decade (2014-2024), offering insights into the state's exposure to such events. The natural disaster risk index score provides a comprehensive risk assessment for each state concerning 18 different natural hazards, reflecting the state's vulnerability to natural disasters. The average annual temperature represents the average yearly temperature of a state over a 30-year period (1994-2024), helping identify climatic trends. The average max temperature captures the average of the highest temperatures during the summer months (June to September) over the same period, essential for understanding summer heatwaves. Conversely, the average min temperature records the average of the lowest temperatures during the winter months (November to February), assessing winter severity. Lastly, the seasonal temperature variation variable measures the variation in temperature across different seasons, averaged over 30 years, providing a clear picture of climatic shifts and seasonal extremes affecting various aspects of life and infrastructure.

APPENDIX C. Factors Analysis Regression Results

Call:

```
lm(formula = Military_Retirees ~ Retirement_Tax_Exemption + Real_Per_Capita_Personal_Income +
  Regional_Price_Parities + Population_Density + No_of_Active_Duty_Personnel +
  Construction_Jobs + Real_estate_and_rental_and_leasing +
  Government_and_government_enterprises, data = MDVA)
```

Residuals:

Min	1Q	Median	3Q	Max
-114134	-5460	-595	6001	48551

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.031e+05	2.198e+04	4.690	6.28e-06	***
Retirement_Tax_Exemption	6.490e+03	3.567e+03	1.820	0.07091	.
Real_Per_Capita_Personal_Income	-6.092e-01	2.455e-01	-2.482	0.01422	*
Regional_Price_Parities	-7.057e+02	2.168e+02	-3.255	0.00142	**
Population_Density	3.183e+00	9.893e-01	3.218	0.00159	**
No_of_Active_Duty_Personnel	5.109e-01	4.142e-02	12.332	< 2e-16	***
Construction_Jobs	2.506e-01	2.187e-02	11.460	< 2e-16	***
Real_estate_and_rental_and_leasing	5.760e-02	2.140e-02	2.692	0.00795	**
Government_and_government_enterprises	-1.281e-01	1.223e-02	-10.476	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14980 on 144 degrees of freedom

Multiple R-squared: 0.9062, Adjusted R-squared: 0.901

F-statistic: 174 on 8 and 144 DF, p-value: < 2.2e-16

Technical Notes:

- Used R Studio to perform regression analysis.

- Interpretation of Regression Results:

- Significance Levels

- Most Significant Variables: Number of Active Duty Personnel, Construction Jobs, and Government Jobs. These variables are most significant, with p-values < 2e-16, which is effectively 0%. This indicates an extremely strong relationship with the number of military retirees.
- Highly Significant Variables: Real Estate Jobs, Regional Price Parities, and Population Density. These variables are significant at the 0.1% level of significance (p-value < 0.001), indicating a very strong relationship with the dependent variable.
- Significant Variable: Retirement Tax Exemption. This variable is significant at the 5% level of significance (p-value = 0.07091), suggesting a meaningful, albeit weaker, relationship with the number of military retirees.

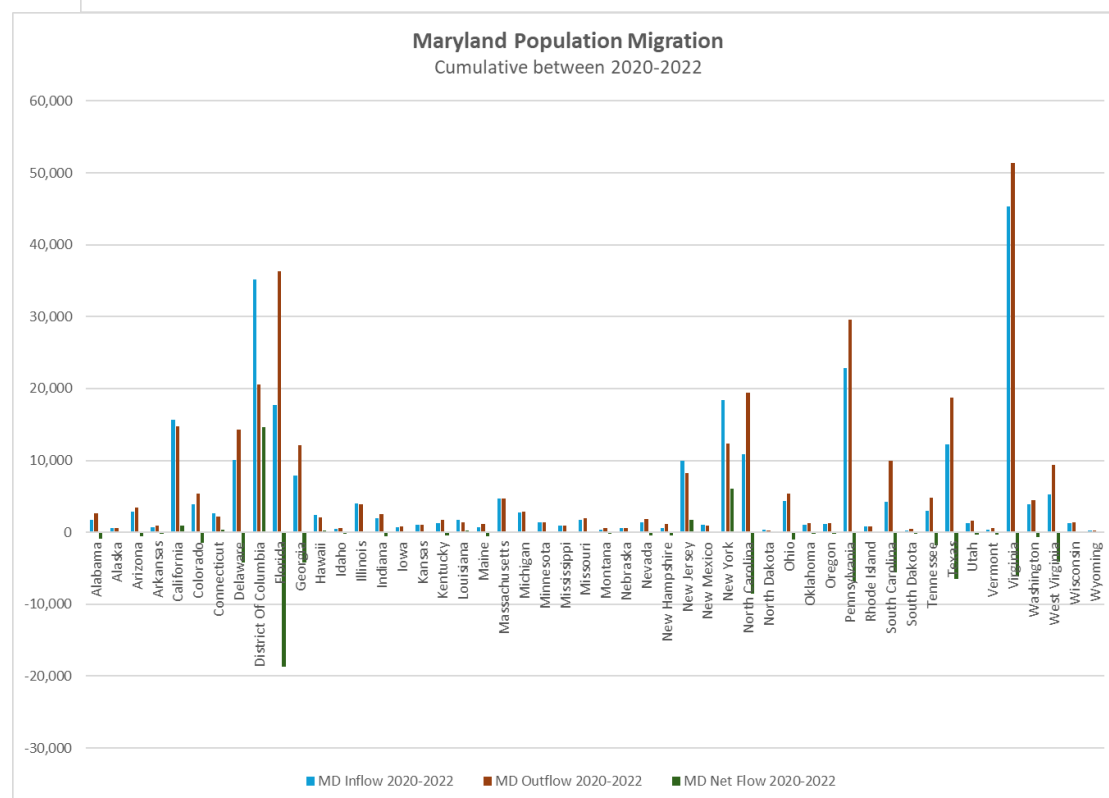
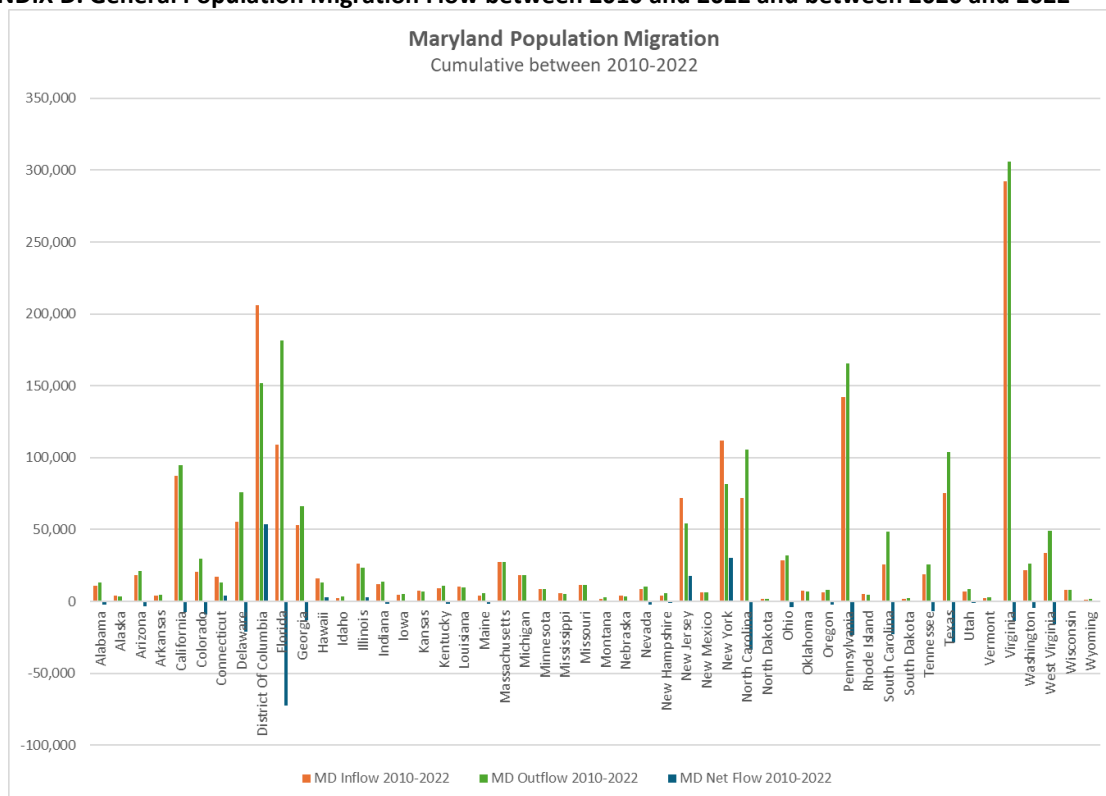
- Income and Price Levels:

- Real Per Capita Personal Income: Higher personal income is associated with fewer military retirees, as indicated by a negative coefficient (-6.092e-01) and significant p-value (0.01422). This suggests that as personal income increases, the number of retirees decreases.
- Regional Price Parities: Higher regional price levels are also associated with fewer military retirees, shown by a negative coefficient (-7.057e+02) and high significance (p-value = 0.00142). This implies that more expensive areas tend to have fewer military retirees.

- Population and Jobs:

- Population Density: Higher population density is positively associated with the number of military retirees (coefficient = $3.183e+00$, p-value = 0.00159). This suggests that more densely populated areas attract more military retirees.
- Construction Jobs: More construction jobs are strongly linked to a higher number of military retirees (coefficient = $2.506e-01$, p-value < $2e-16$). This indicates that areas with more construction activity are attractive to retirees.
- Number of Active Duty Personnel: This variable has a very strong positive relationship with the number of military retirees (coefficient = $5.109e-01$, p-value < $2e-16$). This strongly suggest that areas with a higher military presence tend to attract more retirees.
- Real Estate and Rental Leasing Jobs: More activity in real estate and rental businesses is associated with more retirees (coefficient = $5.760e-02$, p-value = 0.00795). This suggests that areas with vibrant real estate markets which produces employment are attractive to retirees.
- Government and Government Enterprises: More government and government enterprise activity is associated with fewer military retirees (coefficient = $-1.281e-01$, p-value < $2e-16$). This indicates that areas with higher government job activity might be less attractive to military retirees.
- Model Fit:
 - Residual Standard Error: 14980 on 144 degrees of freedom, indicating the average distance between observed and predicted values.
 - Multiple R-squared: 0.9062, meaning that approximately 90.62% of the variability in the number of military retirees can be explained by the model.
 - Adjusted R-squared: 0.901, a slightly adjusted measure that accounts for the number of predictors in the model.
 - F-statistic: 174 on 8 and 144 DF, with a p-value < $2.2e-16$, confirming that the overall model is statistically significant.
- Summary: The regression analysis indicates that several factors significantly influence the number of military retirees in a state. The number of active-duty personnel, construction jobs, and government jobs are the most significant predictors. Higher personal income and regional prices are associated with fewer retirees, while higher population density and more construction jobs attract more retirees. The presence of active-duty personnel strongly correlates with more retirees, and real estate activity positively influences retiree numbers, while government job activity negatively influences them. The high R-squared value suggests the model explains a large portion of the variance in the dependent variable, and the F-statistic confirms the model's overall significance.

APPENDIX D. General Population Migration Flow between 2010 and 2022 and between 2020 and 2022

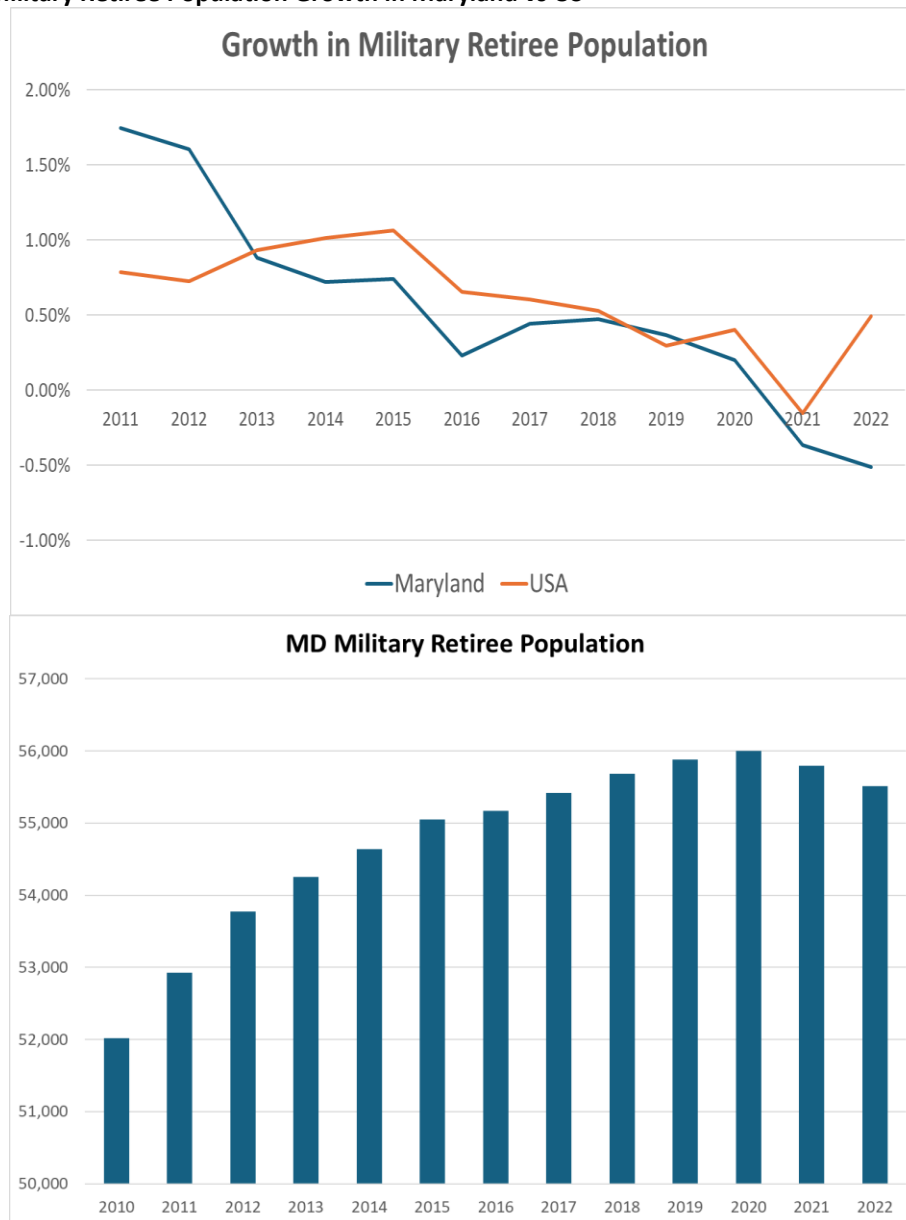


Data Source: IRS US Population Migration Data. <https://www.irs.gov/statistics/soi-tax-stats-migration-data>. Accessed July 2024.

General observations:

- States with high inflow to Maryland: Virginia, followed by Pennsylvania, New York, and North Carolina, have the highest inflow of people to Maryland. This suggests a strong population movement from neighboring states.
- States with high outflow from Maryland: California, Florida, and Texas have the highest outflow of people from Maryland. This could indicate factors like job opportunities, lifestyle preferences, or retirement destinations influencing migration patterns.
- States with significant net inflow to Maryland: Virginia, Pennsylvania, and New York have positive net flows, indicating a net gain of population for Maryland from these states.
- States with significant net outflow from Maryland: California, Florida, and Texas have negative net flows, indicating a net loss of population for Maryland to these states.

APPENDIX E. Military Retiree Population Growth in Maryland vs US



Data Source: Statistical Reports on the Military Retirement System for Each of the Fiscal Years Ended 2010 to 2022. US Department of Defense. <https://actuary.defense.gov/>. Accessed July 2024.

General observations:

- **Slower Growth:** The military retiree population in Maryland has experienced slower growth compared to the national average.
- **Consistent Growth:** Despite the slower pace, there has been consistent growth in the Maryland military retiree population over the analyzed period. The data suggests a slowdown in the growth rate of Maryland's military retiree population after 2020.

APPENDIX F. Regional Military Retiree Population Movements Post-Pandemic

	State w/ Full Exemption on Military Retirement Income (2022)	Δ Military Retirees (2020 vs 2022)	% Change in Military Retiree (2020 vs 2022)	Δ Total Population (2020 vs 2022)	% Change in Total Population (2020 vs 2022)
East North Central	5	1,285	0.78%	-270,223	-0.57%
East South Central	3	2,523	1.42%	171,136	0.88%
Middle Atlantic	3	-1,317	-1.18%	-586,835	-1.38%
Mountain	4	254	0.12%	603,209	2.42%
New England	5	-666	-1.13%	4,534	0.03%
Pacific	3	-11,041	-4.02%	-432,275	-0.81%
South Atlantic	4	-145,063	-20.45%	1,355,380	2.05%
South Atlantic (w/o Va)	4	12,403	2.25%	1,307,674	2.28%
West North Central	7	898	0.72%	67,797	0.31%
South West (4 state)	4	5,323	1.72%	909,407	2.23%
Whole USA	38	-147,804	-6.90%	1,822,130	0.55%
Maryland	0	-489	-0.87%	-13,243	-0.21%

Data Sources:

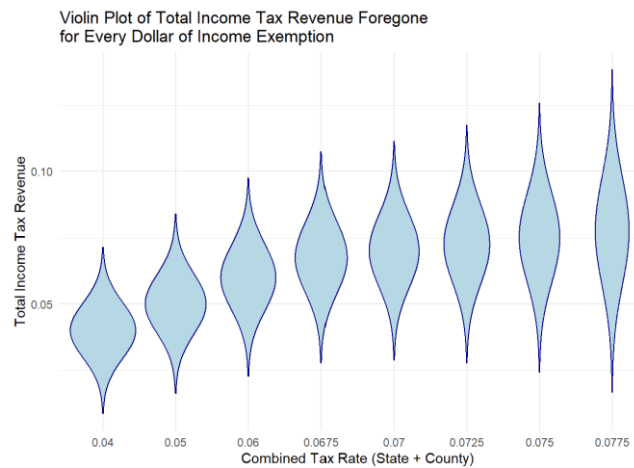
- United States Census Bureau. <https://data.census.gov/>. Accessed July 2024.
- Statistical Reports on the Military Retirement System for Fiscal Years Ended September 30, 2022 and September 30, 2020. US Department of Defense. <https://actuary.defense.gov/>. Accessed July 2024.

General observations:

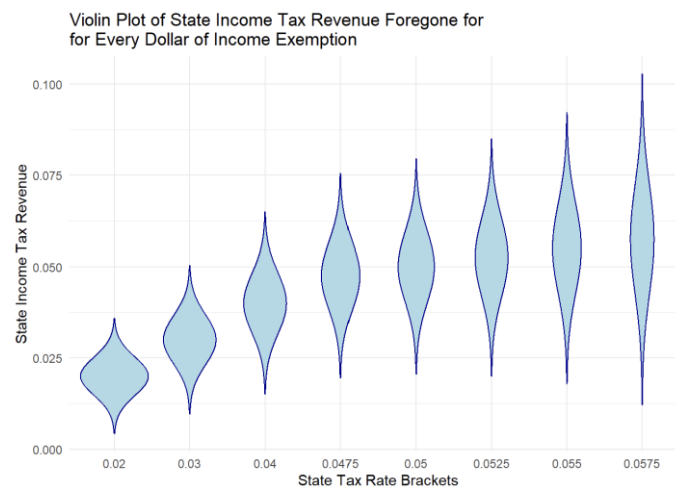
- Overall Trends:
 - National Decline in Military Retirees: There's a significant overall decrease of 6.90% in the military retiree population between 2020 and 2022.
 - Population Growth: The total population increased by 0.55% nationally during the same period.
- Regional Trends:
 - Southward Migration: The South Atlantic region (excluding Virginia) saw a substantial increase in both military retirees (2.25%) and total population (2.28%).
 - Westward Migration: The Mountain and West South Central regions experienced growth in both military retirees and total population.
 - East Coast Losses: The Middle Atlantic and New England regions lost military retirees, with the Middle Atlantic also experiencing a decline in total population.
- State-Level Impact:
 - Maryland: Experienced a slight decline in both military retirees (-0.87%) and total population (-0.21%).
 - Virginia's Significant Decline: Virginia experienced the most substantial loss of military retirees, exceeding 157,000.
 - Beneficiary States: Southern Atlantic states (excluding DC and Maryland) and Texas gained military retirees during this period.

APPENDIX G. Potential Foregone Marginal Income Tax from Existing Military Retiree Residing in Maryland

Results for Total Income Tax = State Tax + Local Tax



Results for State Tax



Technical Notes:

- R Studio was used as a tool in performing the estimation using Monte Carlo.

Assumptions:

- Local income tax is fixed at 3.2% (range is between 2.25%-3.2%).
- Existing state income tax rates and brackets as of August 2024 are followed.
 - Distribution per tax rate and brackets following military retiree are taxpayers filing joint returns, head of household, or qualifying widows/widowers. The distribution used is indicated below:

<i>Taxable Net Income</i>	<i>State Tax</i>	<i>Sample Distribution</i>
\$0 - \$1,000	2.00%	5%
\$1,000 - \$2,000	\$20 plus 3.00% of the excess over \$1,000	10%
\$2,000 - \$3,000	\$50 plus 4.00% of the excess over \$2,000	15%
\$3,000 - \$150,000	\$90 plus 4.75% of the excess over \$3,000	20%
\$150,000 - \$175,000	\$7,072.50 plus 5.00% of the excess over \$150,000	20%
\$175,000 - \$225,000	\$8,322.50 plus 5.25% of the excess over \$175,000	15%
\$225,000 - \$300,000	\$10,947.50 plus 5.50% of the excess over \$225,000	10%
Over \$300,000	\$15,072.50 plus 5.75% of the excess over \$300,000	5%

- Average Total Tax takes into account both state income tax and local income tax. Local tax rate is fixed at 3.2%.

Descriptive Summary Statistics:

- Total Income Tax per Military Retiree per Dollar Exemption
 - Mean: \$0.0658155. This is the average total income tax that a military retiree pays per dollar exemption. On average, each dollar exemption corresponds to approximately \$0.066 in total income tax.
 - Median: \$0.07. This indicates that the median total income tax per military retiree per dollar exemption is \$0.07. This value represents the middle point of the data, where half of the values are below \$0.07 and half are above.
 - Standard Deviation: \$0.009365027. The standard deviation measures the dispersion or variability around the mean. A standard deviation of approximately \$0.0094 indicates that the total income tax per military retiree per dollar exemption typically varies by this amount from the mean value of \$0.0658.
 - 90% Confidence Interval: \$0.05 to \$0.075. This confidence interval suggests that we can be 90% confident that the true mean total income tax per military retiree per dollar exemption lies between \$0.05 and \$0.075. This interval provides a range within which the true mean is likely to fall, with 90% confidence.
- State Income Tax per Military Retiree per Dollar Exemption
 - Mean: \$0.0458155. This is the average state income tax that a military retiree pays per dollar exemption. On average, each dollar exemption corresponds to approximately \$0.046 in state income tax.
 - Median: \$0.05. The median state income tax per military retiree per dollar exemption is \$0.05. This value represents the middle point of the data, where half of the values are below \$0.05 and half are above.
 - Standard Deviation: \$0.009365027. The standard deviation measures the dispersion or variability around the mean. A standard deviation of approximately \$0.0094 indicates that the state income tax per military retiree per dollar exemption typically varies by this amount from the mean value of \$0.0458.
 - 90% Confidence Interval: \$0.03 to \$0.055. This confidence interval suggests that we can be 90% confident that the true mean state income tax per military retiree per dollar exemption lies between \$0.03 and \$0.055. This interval provides a range within which the true mean is likely to fall, with 90% confidence.

APPENDIX H. Descriptive Summary Statistics for Select Jobs' Household Income, Total Income Tax, and State Income Tax

Income per Job	Mean_Household_Income	Median_Household_Income	Std_Dev_Household_Income	CI_90_Lower_Household_Income	CI_90_Upper_Household_Income
Construction Managers	137,994	136,123	50,827	57,580	225,668
Construction and Extraction Occupations	76,908	71,384	32,317	31,454	136,637
Defense Contracting	97,025	94,855	43,348	28,727	171,224
First-Line Supervisors of Construction Trades and Extraction Workers	96,393	92,245	35,207	44,658	160,967
Government Employees	68,617	58,458	26,615	38,943	120,188
Real Estate Agents	85,315	81,937	41,459	21,271	159,977
Real Estate Managers	106,218	104,262	40,908	42,134	176,996
Overall	95,496	90,188	44,629	35,228	176,535
Total Income Tax per Job	Mean_Total_Tax	Median_Total_Tax	Std_Dev_Total_Tax	CI_90_Lower_Total_Tax	CI_90_Upper_Total_Tax
Construction Managers	10,977	10,769	4,137	4,525	18,206
Construction and Extraction Occupations	6,062	5,622	2,570	2,448	10,810
Defense Contracting	7,670	7,488	3,467	2,231	13,613
First-Line Supervisors of Construction Trades and Extraction Workers	7,615	7,281	2,808	3,498	12,772
Government Employees	5,403	4,595	2,116	3,043	9,502
Real Estate Agents	6,735	6,462	3,308	1,639	12,691
Real Estate Managers	8,403	8,236	3,277	3,297	14,091
Overall	7,552	7,117	3,581	2,748	14,052
State Tax per Job	Mean_State_Tax	Median_State_Tax	Std_Dev_State_Tax	CI_90_Lower_State_Tax	CI_90_Upper_State_Tax
Construction Managers	6,561	6,413	2,511	2,683	10,984
Construction and Extraction Occupations	3,601	3,338	1,536	1,442	6,438
Defense Contracting	4,565	4,453	2,080	1,312	8,134
First-Line Supervisors of Construction Trades and Extraction Workers	4,530	4,329	1,682	2,069	7,621
Government Employees	3,207	2,724	1,264	1,797	5,656
Real Estate Agents	4,005	3,840	1,981	958	7,571
Real Estate Managers	5,004	4,900	1,968	1,949	8,427
Overall	4,496	4,231	2,153	1,621	8,403

Data Sources:

- Maryland Income Tax Rates and Brackets. [Tax Rates \(marylandtaxes.gov\)](https://taxrates.marylandtaxes.gov/). Accessed July 2024.
- May 2023 State Occupational Employment and Wage Estimates for Maryland. US Bureau of Labor Statistics. [Maryland - May 2023 OEWS State Occupational Employment and Wage Estimates \(bls.gov\)](https://www.bls.gov/news.release/may2023oeaws.htm). Accessed July 2024.
- Department of Defense Contractor Salary in Maryland. Zip Recruiter. <https://www.ziprecruiter.com/Salaries/Dod-Contractor-Salary--in-Maryland#TableView>. Accessed July 2024.

Technical Note:

- R Studio was used as a tool in performing the estimation using Monte Carlo.

Assumptions:

- Existing state income tax rates and brackets as of August 2024 are followed.
 - Distribution per tax rate and brackets following military retiree are taxpayers filing joint returns, head of household, or qualifying widows/widowers. The distribution used is indicated below:

Taxable Net Income	State Tax	Sample Distribution
\$0 - \$1,000	2.00%	5%
\$1,000 - \$2,000	\$20 plus 3.00% of the excess over \$1,000	10%
\$2,000 - \$3,000	\$50 plus 4.00% of the excess over \$2,000	15%
\$3,000 - \$150,000	\$90 plus 4.75% of the excess over \$3,000	20%
\$150,000 - \$175,000	\$7,072.50 plus 5.00% of the excess over \$150,000	20%
\$175,000 - \$225,000	\$8,322.50 plus 5.25% of the excess over \$175,000	15%
\$225,000 - \$300,000	\$10,947.50 plus 5.50% of the excess over \$225,000	10%
Over \$300,000	\$15,072.50 plus 5.75% of the excess over \$300,000	5%

- Marital status and the corresponding employment status of spouses are assumed to be have below distribution based on US Census Bureau and US Government Accountability Office²².

Not Married	40%
Married; Spouse Not Working	30%
Married; Spouse Working Parttime	10.20%
Married; Spouse Working Full time	19.80%

- Below is the assumed spouse income²³ based on 90% of them are female.

Military Spouse	Annual Salary	Std Deviation
Not Working	-	-
Part Time Work	37,863	4,583
Full Time Work	60,580	4,583

- Salary is treated as uncertain input in the simulation. This variable is assumed normal distribution while standard deviation is calculated using interquartile range from raw data sourced from Bureau of Labor Statistics for May 2023.
- Local income tax is fixed at 3.2%.
- Average Total Tax takes into account both state income tax and local income tax.

²² US Government Accountability Office. [Military Spouse Employment: Part-Time Workforce Characteristics and Perspectives \[Reissued with revisions on May 9, 2024\] | U.S. GAO](#). Accessed August 2024.

²³ Women's Earnings in Maryland 2022. Bureau of Labor Statistics. [Women's Earnings in Maryland — 2022 : Mid-Atlantic Information Office : U.S. Bureau of Labor Statistics \(bls.gov\)](#). Accessed August 2024.

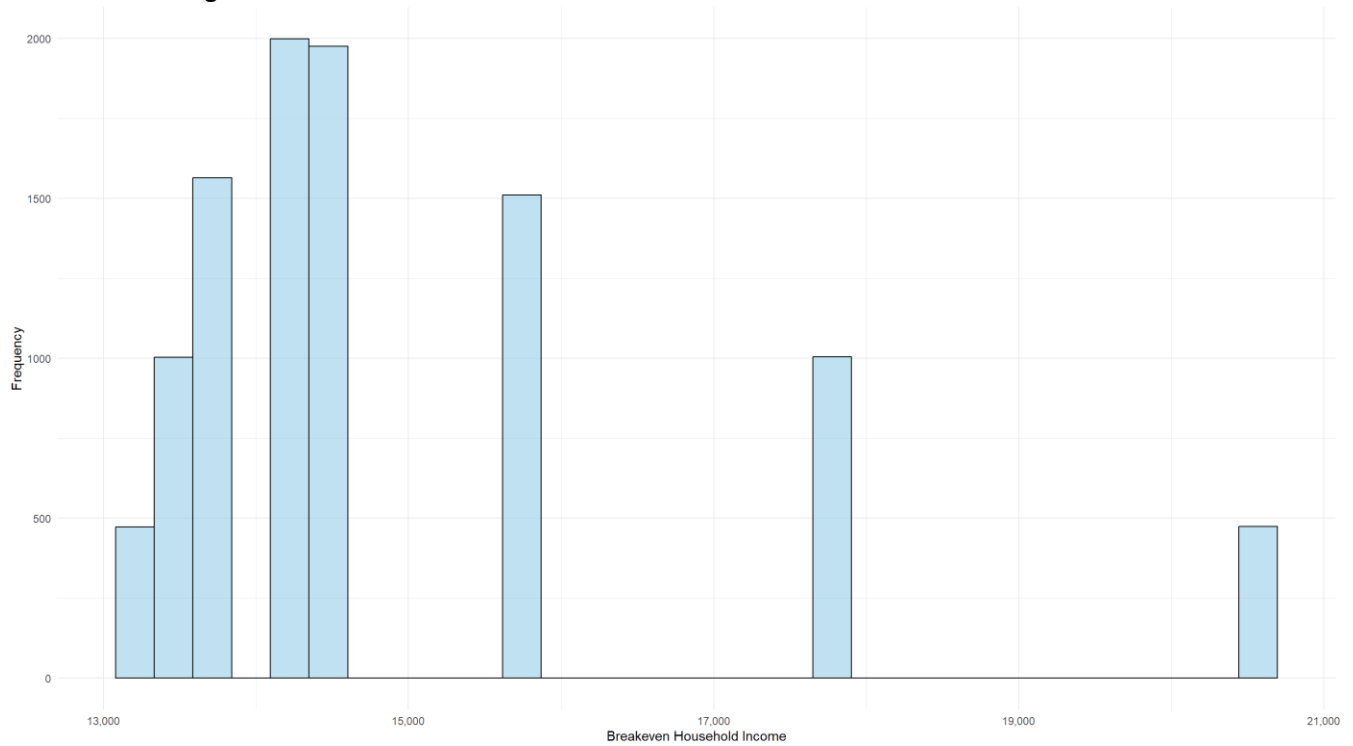
APPENDIX I. Breakeven Income for Every Dollar of Exemption

Breakeven Income for Every Dollar of Exemption		Tax Brackets of Existing Retiree							
		2%	3%	4%	4.75%	5%	5.25%	5.50%	5.75%
Tax Bracket for New Military Retiree or Higher Income for Existing Retiree	2%	0.94	1.41	1.88	2.23	2.35	2.47	2.59	2.70
	3%	0.63	0.94	1.25	1.49	1.57	1.65	1.72	1.80
	4%	0.47	0.71	0.94	1.12	1.18	1.23	1.29	1.35
	4.75%	0.40	0.59	0.79	0.94	0.99	1.04	1.09	1.14
	5%	0.38	0.56	0.75	0.89	0.94	0.99	1.03	1.08
	5.25%	0.36	0.54	0.72	0.85	0.90	0.94	0.98	1.03
	5.50%	0.34	0.51	0.68	0.81	0.85	0.90	0.94	0.98
	5.75%	0.33	0.49	0.65	0.78	0.82	0.86	0.90	0.94

Computation:

Breakeven Income for Every Dollar of Exemption = [Tax Rate of Existing Retiree – (Tax Rate of Existing Retiree x Sales Tax)] / Tax Rate for New Retiree

APPENDIX J. Histogram of Simulated Household Income



Mean Breakeven Household Income: \$ 14,954.58

90% Confidence Interval: \$ 13,505.46 to \$ 17,741.68

Technical Note:

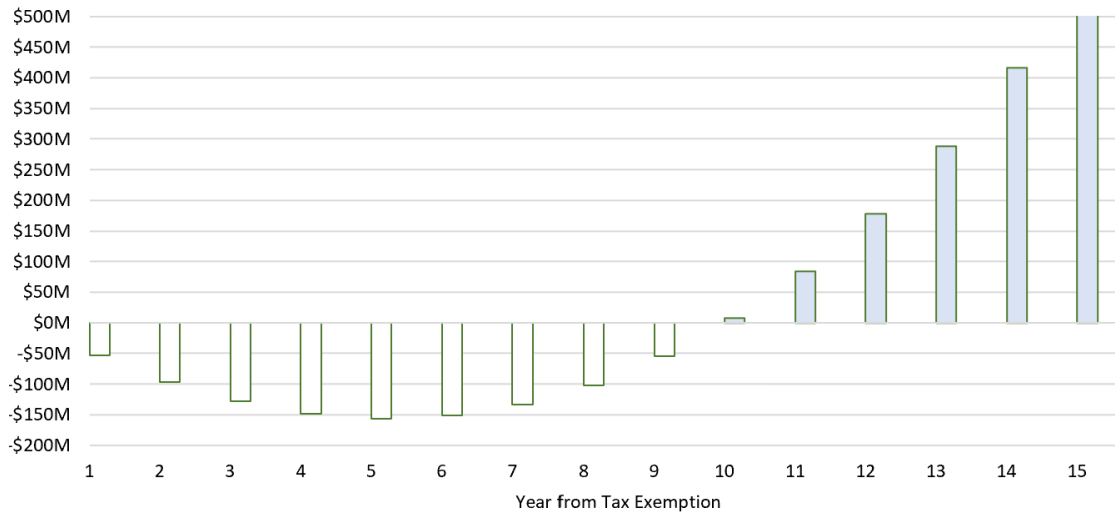
- R Studio was used as a tool in performing the estimation using Monte Carlo.

Assumptions:

- Total Foregone Tax for Full Exemption = \$61.3 Million (from SB 349)
- Existing Military Retirees in Maryland = 55.5K (from DoD 2022 military retirees in Maryland)
- Estimated Foregone Income Tax per Household for Full Exemption = \$1100
- Local income tax rate is fixed 3.2%.
- Existing state income tax rates and brackets as of August 2024 are followed.
 - Distribution per tax rate and brackets following military retiree are taxpayers filing joint returns, head of household, or qualifying widows/widowers. The distribution used is indicated below:

Taxable Net Income	State Tax	Sample Distribution
\$0 - \$1,000	2.00%	5%
\$1,000 - \$2,000	\$20 plus 3.00% of the excess over \$1,000	10%
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\$3,000 - \$150,000	\$90 plus 4.75% of the excess over \$3,000	20%
\$150,000 - \$175,000	\$7,072.50 plus 5.00% of the excess over \$150,000	20%
\$175,000 - \$225,000	\$8,322.50 plus 5.25% of the excess over \$175,000	15%
\$225,000 - \$300,000	\$10,947.50 plus 5.50% of the excess over \$225,000	10%
Over \$300,000	\$15,072.50 plus 5.75% of the excess over \$300,000	5%

APPENDIX K. Net Income Tax Revenue Forecast



Data Sources:

- United States Census Bureau. <https://data.census.gov/>. Accessed July 2024.
- Office of the Actuary US Department of Defense. <https://actuary.defense.gov/>. Accessed July 2024.
- Military One Source. <https://www.militaryonesource.mil/data-research-and-statistics/military-community-demographics/>. Accessed July 2024.

Assumptions / Intermediate Calculations:

- Estimated average retirement pay person for retirees who are in Maryland:

Rank & Age	Avg Retirement Pay Per Person	Count
Officer	\$ 55,686	16,163
35-44	\$ 43,702	323
45-54	\$ 51,151	1,778
55-64	\$ 56,125	3,717
>65	\$ 56,683	10,344
Enlisted	\$ 26,040	32,375
35-44	\$ 24,952	1,295
45-54	\$ 26,143	5,180
55-64	\$ 25,491	9,389
>65	\$ 26,405	16,511
Total	\$ 35,823	48,538

- Estimated income tax revenue foregone:

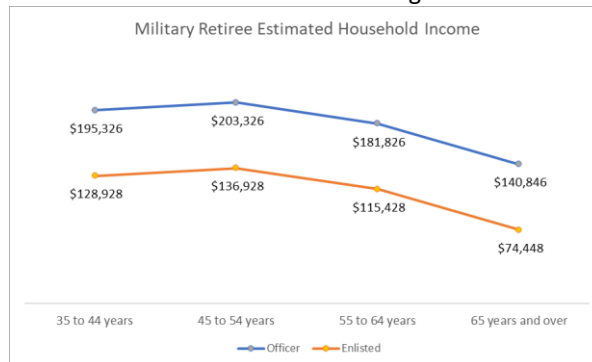
Type & Age	Local Income Tax Revenue	State Income Tax Revenue
Officer	\$ 18,961,886	\$ 28,675,612
35-44	\$ 322,767	\$ 518,729
45-54	\$ 2,199,023	\$ 3,550,767
55-64	\$ 4,297,467	\$ 6,581,901
>65	\$ 12,142,629	\$ 18,024,215
Enlisted	\$ 7,811,536	\$ 11,595,249
35-44	\$ 516,026	\$ 765,976
45-54	\$ 2,261,495	\$ 3,356,906
55-64	\$ 1,649,771	\$ 2,448,879
>65	\$ 3,384,244	\$ 5,023,488
Total	\$ 26,773,422	\$ 40,270,861
Grand Total		\$ 67,044,283

- Other than income taxes, the following were considered:
 - Sales tax revenue estimated at 6% tax rate at \$29,158 (Per Capita Personal Consumption Expenditures in MD,2022 from US Bureau of Economic Analysis).
 - Gasoline tax revenue estimated at 14.5% tax rate at \$1,401.
 - Average assessed value of residential property in Maryland is estimated at \$268,443 (from Maryland Department of Assessments & Taxation 2021 Ratio Report) with 1.1% property tax rate. It is further assumed that 75% of military retirees are homeowners.

- Estimated annual taxes collected, other than income tax:

Sales Tax	Gasoline Tax	Property Tax	Total
\$ 2,834.16	\$ 329.09	\$ 2,214.65	\$ 5,377.91

- Assumed the following household income based on rank and age brackets.



- From US Mean Household Income of \$106.4K, the estimates above were adjusted to reflect assumptions on income-based family/marital status, educational attainment, house rental status, Mid-Atlantic region premium, and age.
- Maryland's general population median household income is \$95K as of 2022, according to the US Bureau of Census. Due to the likely experience and premium on military retirees' labor contribution, the estimated household income is projected to be in the upper quartile for officers and above the median for enlisted personnel.